Polynomial Input Preconditioning for Zero-Shot Time-Series Forecasting

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Problem

Patch-based time-series transformers embed each patch independently before attention. This is efficient, but it means each initial token only sees its local P-step window. Long-range structure across patches has to be discovered entirely inside the transformer.
Research question: can a fixed polynomial transformation expose useful cross-patch structure before tokenization?

Preliminaries

Universal Sequence Preconditioning preprocesses a sequence with a fixed polynomial convolution,

$$x_t^{\text{pre}} = \sum_{k=0}^{d} c_k x_{t-k}.$$

In the online linear-dynamical-systems setting, this kind of transformation can improve regret bounds by reducing the effective difficulty of the hidden dynamics. This project tests whether the same idea helps in a different setting: offline pretraining and zero-shot multi-step forecasting.

Method

For a normalized time series x_t, compute a fixed Chebyshev residual

$$r_t = \sum_{k=1}^{d} c_k x_{t-ks}.$$

With $d = 4$ and stride $s = P = 16$,

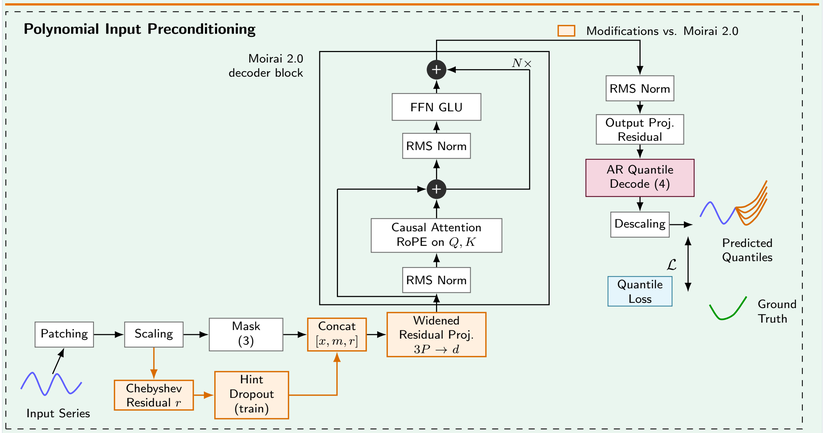
$$r_t = -x_{t-32} + \frac{1}{8} x_{t-64}.$$

Concatenate this as a third input channel:

$$\mathbf{e}_i = \text{ResBlock}_{3P\to d}([\mathbf{p}_i; \mathbf{m}_i; \mathbf{r}_i]).$$

Only the input projection widens; the transformer decoder and forecasting target are unchanged.

Architecture



Orange components are the only changes relative to Moirai 2.0: preconditioning residual, optional training-time channel dropout, concatenation, and the widened input projection.

Experimental Design

- Model: Moirai 2.0 Small, 11.4M parameters.
- Data: public LOTSA setup, 27B observations.
- Main budget: 10K optimizer steps, 5 seeds.
- Extended budget: 100K optimizer steps, 5 seeds.
- Primary benchmark: GIFT-Eval, 97 dataset-by-horizon configurations.
- External benchmark: FEV-Bench, 100 forecasting tasks.
- Metric: geometric-mean MASE; lower is better.

Main Results

The polynomial channel improves GIFT-Eval normalized MASE by 2.9%, wins 72 of 97 configurations after averaging across five seeds, adds only 0.11% parameters, and reaches a 3.9% improvement at 100K training steps.

Method	MASE	Δ	Wins
`d4_dropout`	**0.837**	−2.9%	**72/97**
`d4`	0.838	−2.8%	72/97
Zero ctrl	0.858	−0.5%	54/97
Baseline	0.862	—	—
Duplicate ctrl	0.865	+0.4%	39/97

The capacity-matched controls show the effect is not just from adding parameters. A zero channel is neutral, while duplicating the raw input hurts.

Statistical Evidence

For GIFT-Eval, I average MASE per configuration across five seeds and run a two-sided paired sign test. `d4_dropout` wins 72 of 97 configurations over the baseline, giving $p < 10^{-5}$.
This is a controlled paired comparison: every model is trained with the same data, optimizer, precision, and seed set within the compared condition.

Fixed vs. Learned Filters

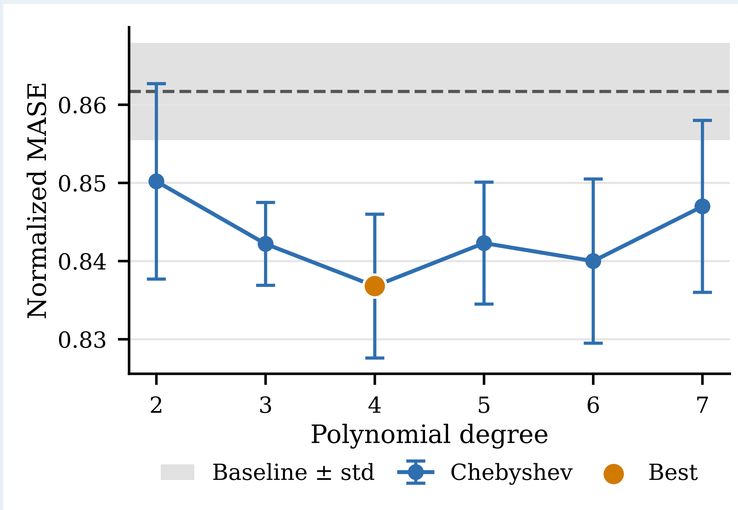
Setting	MASE	Δ
Fixed Chebyshev	**0.8234**	−4.99%
Learned, Cheb. init	0.8332	−3.85%
Learned, zero init	0.8358	−3.57%

The fixed Chebyshev coefficients outperform learned coefficient vectors, even when the learned model is initialized at Chebyshev. This suggests the polynomial channel is a useful inductive bias.

Personal Contribution

- Implemented the preconditioning channel in the Moirai 2.0 training/evaluation pipeline.
- Designed the zero and duplicate capacity controls.
- Ran the seed, degree, coefficient-learning, horizon, and 100K-step analyses.
- Prepared the CSML paper, poster, and presentation materials.
- Thanks to Alex Kawaja for working together on this collaborative project.

Ablations



All tested Chebyshev degrees improve over baseline. Degree 4 is a strong default, not a sharply tuned optimum.

Independent Benchmark

Method	MASE	SQL
`d4_dropout`	**1.2532**	**1.0236**
Baseline	1.2908	1.0562
Δ	−2.91%	−3.09%

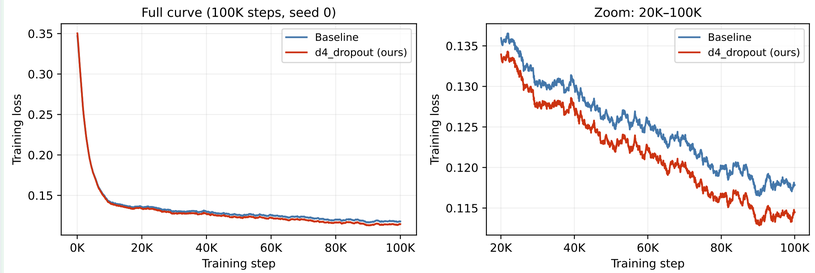
The improvement transfers to FEV-Bench, an independent benchmark with 100 forecasting tasks from 96 datasets.

Horizon Effects

Term	Baseline	`d4`	`d4_dropout`
Short	0.840	**0.825**	0.828
Medium	0.859	0.830	**0.823**
Long	0.924	0.880	**0.874**

The largest gain is on long-horizon tasks: `d4_dropout` improves MASE by 5.3% relative to baseline in the long-horizon bin.

Training Dynamics



At 100K training steps, the MASE improvement grows to −3.9%. The preconditioned model also has lower cross-seed standard deviation: 0.0079 vs 0.0189 for the baseline.

Takeaway: A fixed, theory-guided input channel improves a modern time-series foundation model without changing the transformer stack. The gain comes from polynomial content, not added width, and is strongest at longer training budgets and longer horizons.